



Behavioral
Data Science

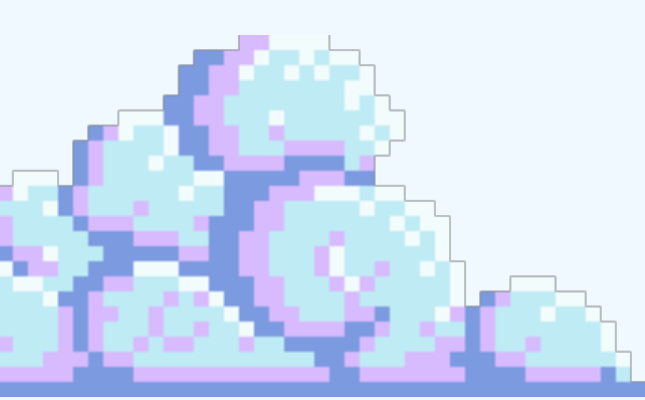
MARIO @ EXIST 25

A SIMPLE GATEWAY TO EFFECTIVE
MULTILINGUAL SEXISM DETECTION

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TASK 1



1

Binary Sexism Identification

Sexist vs. Non-sexist



2

Source Intention Detection

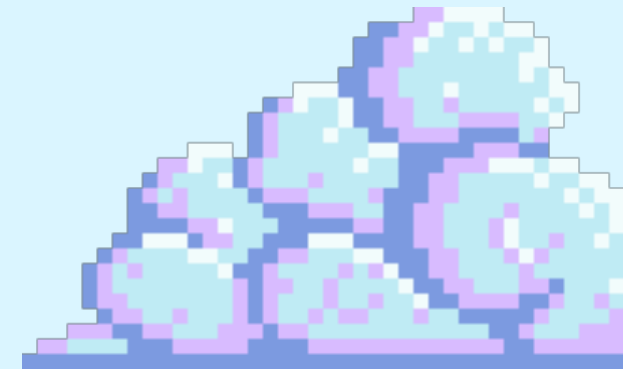
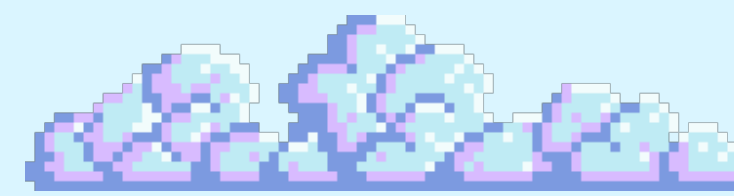
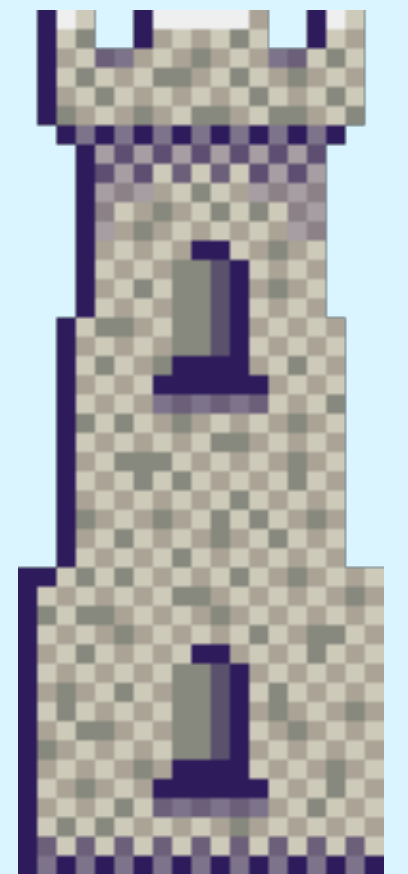
Direct, Reported, Judgemental

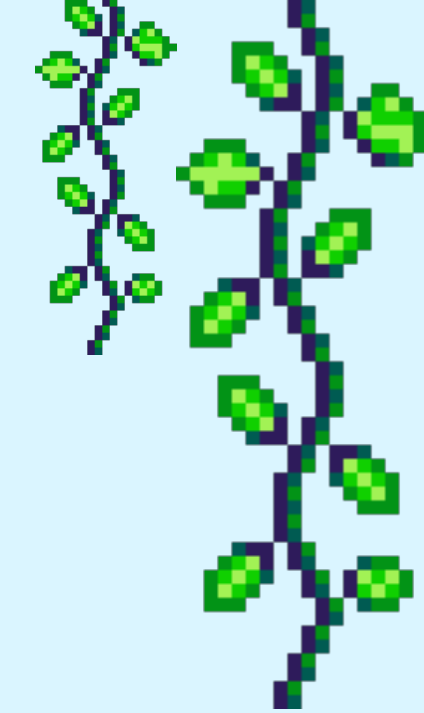
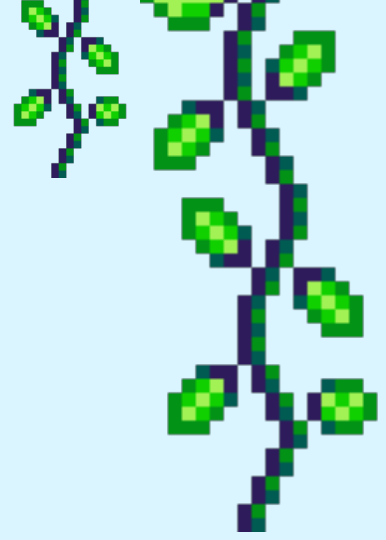


3

Sexism Categorization

Ideological and inequality, stereotyping and dominance, objectification, sexual violence, misogyny and non sexual violence





OUR APPROACH

A unified multilingual model with task-specific adapters and conditional specialization based on the hierarchical label structure.



The background is a pixel art landscape. At the top, there are several pixelated clouds in shades of blue and purple. A yellow five-pointed star with a black outline is positioned in the upper right. A bright yellow sun with a black outline is in the top right corner. On the right side, there is a large, dark green pixelated tree. In the bottom left, a boy with green hair, wearing an orange shirt and blue overalls, stands next to a small brown and white dog. The ground is covered in green grass with small blue and pink flowers. Two small yellow sun-like icons are on the ground in the lower right.

BASE MODEL

We picked Llama 3.1 8B as our base model.
It was pretrained with 8 languages [English,
German, French, Italian, Portuguese, Hindi,
Spanish, and Thai]

OUR SETUP

Data: Gold labels only (no disagreement modeling in this run).

Preprocessing: Minimal (keep hashtags, emojis, etc.)

Training: LoRA fine-tuning with dynamic gating



LORA

Low-Rank Adaptation Fine-tuning of Large Language Models

LoRA

low rank
 $r \ll n$

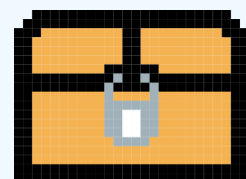
$$\begin{matrix} n \\ \left[\begin{array}{c|c} \text{orange grid} \end{array} \right]_n \end{matrix} + \begin{matrix} \begin{matrix} \left[\begin{array}{c} \text{green grid} \end{array} \right]_r \\ \times \left[\begin{array}{c|c} \text{green grid} \end{array} \right]_r \end{matrix} \end{matrix} \begin{matrix} \left[\begin{array}{c} \text{green grid} \end{array} \right]_n \\ \left[\begin{array}{c} \text{green grid} \end{array} \right]_n \end{matrix}$$

$$W = W_0 + \underline{A} \underline{B}$$

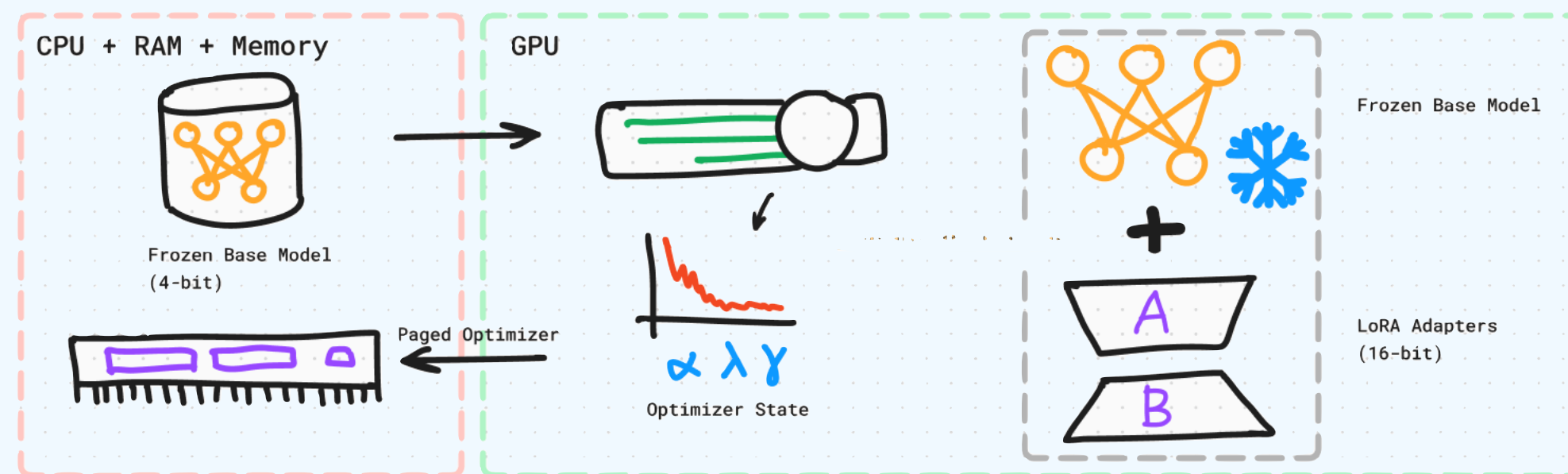
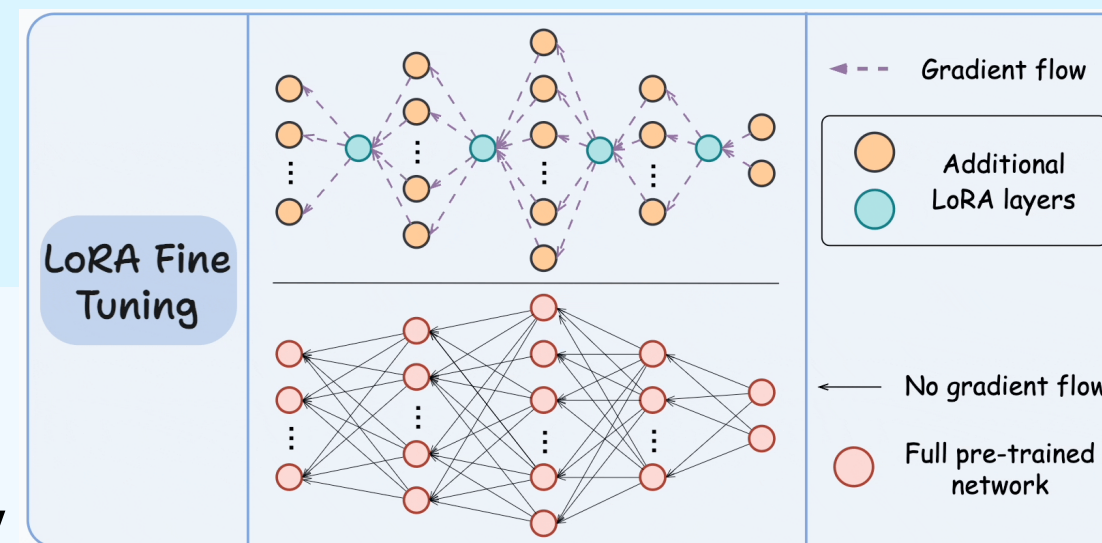
frozen trainable

OUR LORA EXTENSION

TRICKS



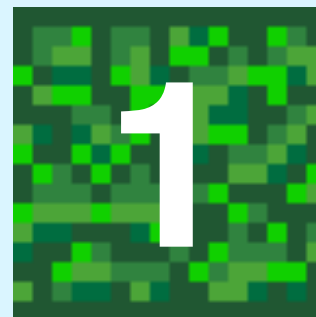
- Why all layers? to capture more patterns than attention-only
- Balance? with quantization using 4-bit QLoRA + Flash Attention v2



TASK LABEL HIERARCHY

Three levels: Binary (1.1) → Intention (1.2) → Categorization (1.3).

Task



Task



Task



Note: Subtasks 1.2 & 1.3 only apply to SEXIST tweets

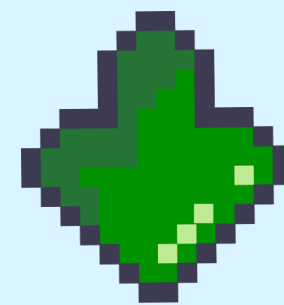
PASSING THE ERROR



Sexist



Intention

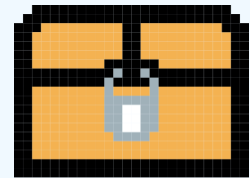


Categorization



HIERARCHICAL LABEL-AWARE ADAPTATION

TRICKS



- During training, adapter routing is conditioned on the gold parent labels.
- During inference, predictions proceed sequentially from the top level to activate the corresponding LoRA module.



LOSS FUNCTIONS

Task-specific + hierarchy constraint
(penalize invalid transitions, e.g., NOT_SEXIST but high category probs).

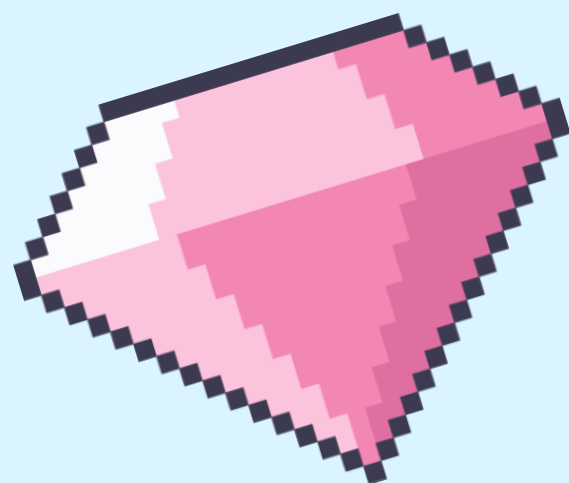


$$\mathbf{h}^{(\ell)} = f_{\theta}(x) + \Delta_{\hat{y}^{(\ell-1)}}^{(\ell)}(x).$$

$$\mathcal{L}_{\text{hierarchy}} = \lambda \sum_{i=1}^N \sum_{\ell=2}^3 \mathbb{F}[\hat{y}_i^{(\ell-1)} = \text{NOT_SEXIST}] \cdot \max_{c \in \mathcal{C}^{(\ell)}} p_i^{(\ell)}(c)$$

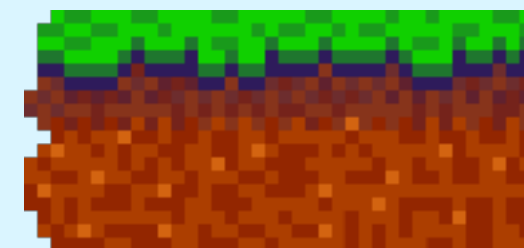
$$\mathcal{L}_{\text{total}} = \sum_{\ell=1}^3 \mathcal{L}_{\text{task}}^{(\ell)} + \mathcal{L}_{\text{hierarchy}}$$





MULTI-LINGUAL

English and Spanish

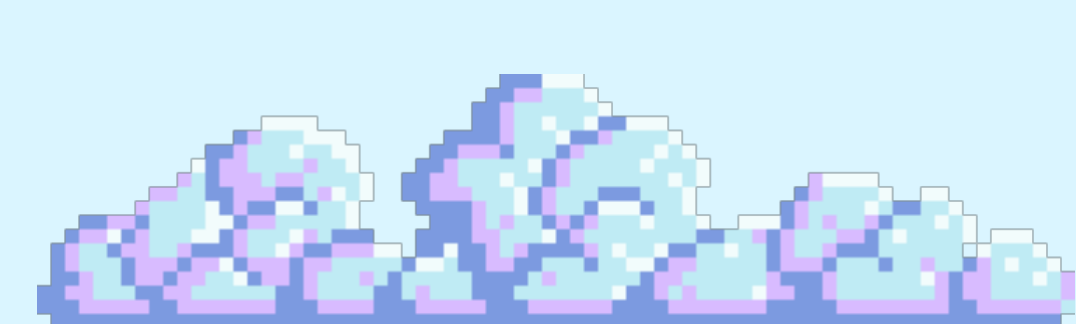
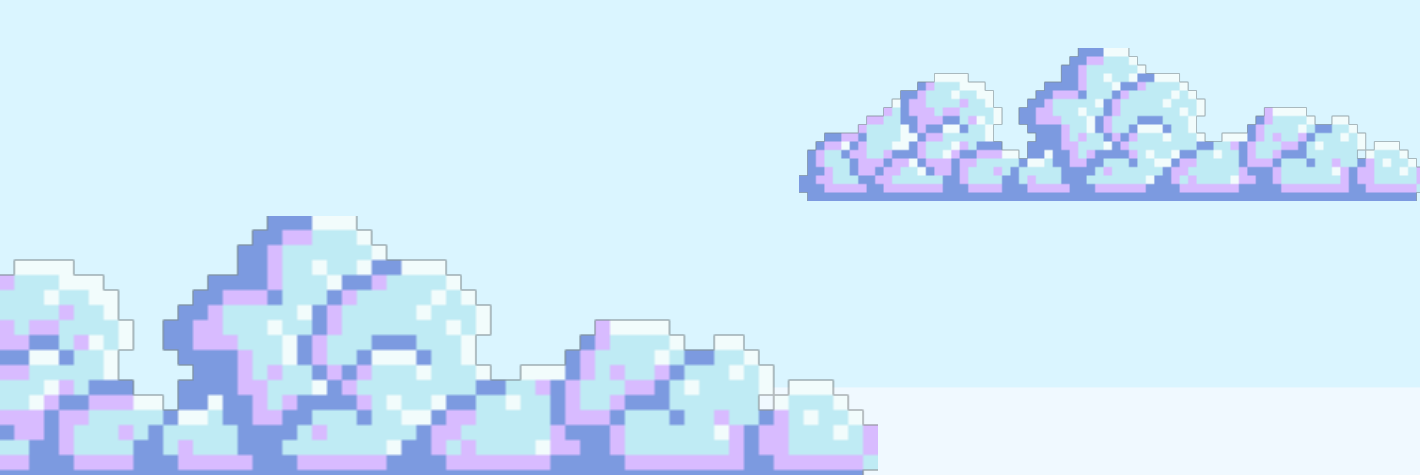


Hello!



Hola!





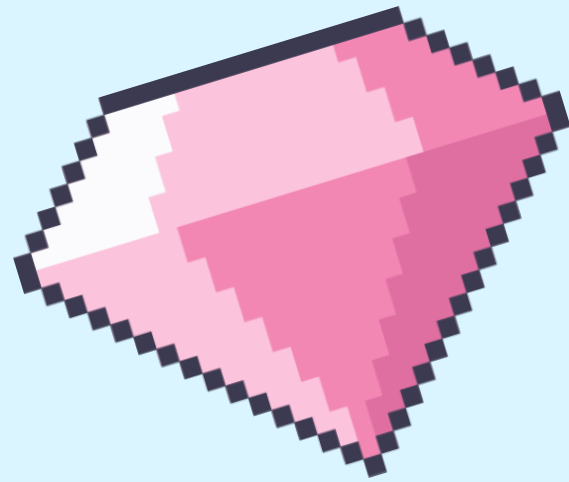
MULTILINGUAL STRATEGY

- 1. Joint training on English + Spanish (no separate models).
- 2. Benefits: Cross-lingual transfer improves both languages symmetrically.
- 3. Training details: Separate adapters per subtask, early stopping on validation.



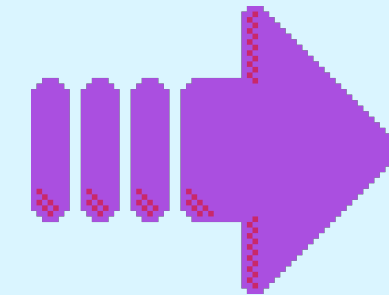
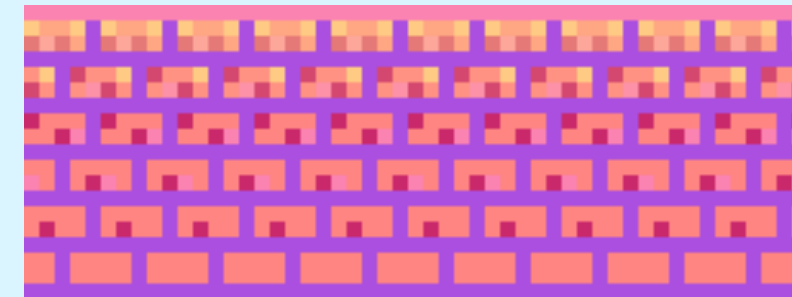
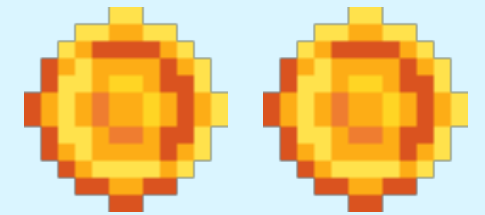
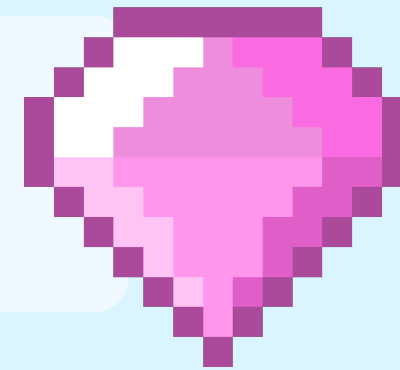
Table 2
Performance Comparison: Joint vs. Separate Language Training (F1 Scores on validation set)

Subtask	Training Strategy	English	Spanish	Average
1.1 (Binary)	Separate	0.847	0.831	0.839
	Joint Training	0.863 (+0.016)	0.851 (+0.020)	0.857 (+0.018)
1.2 (Intention)	Separate	0.742	0.728	0.735
	Joint Training	0.758 (+0.016)	0.745 (+0.017)	0.752 (+0.017)
1.3 (Categorization)	Separate	0.681	0.665	0.673
	Joint Training	0.704 (+0.023)	0.689 (+0.024)	0.697 (+0.024)



SOLUTION

Will this simple
approach work?



TASK-SPECIFIC

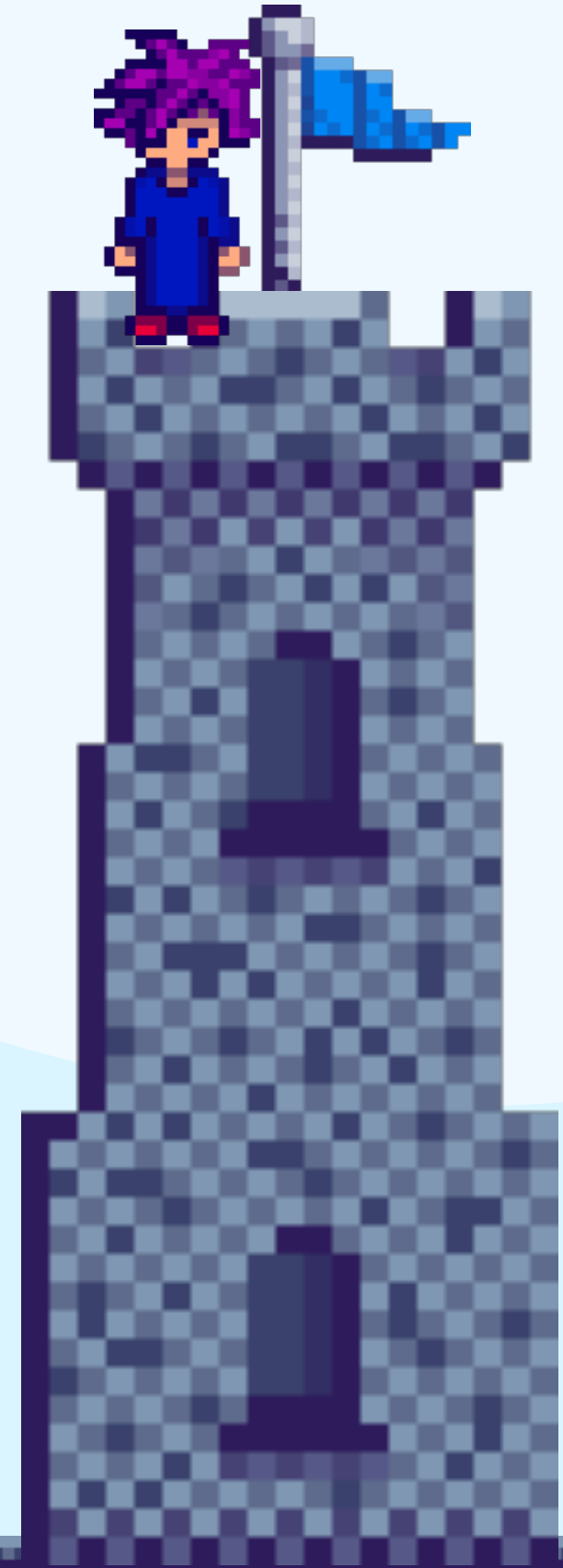
- Binary: 0.6774 (EN: 0.6231, ES: 0.7124)
- Intention: 0.4991 (EN: 0.3676, ES: 0.5937)
- Categorization: 0.6519 (EN: 0.5085, ES: 0.7650)



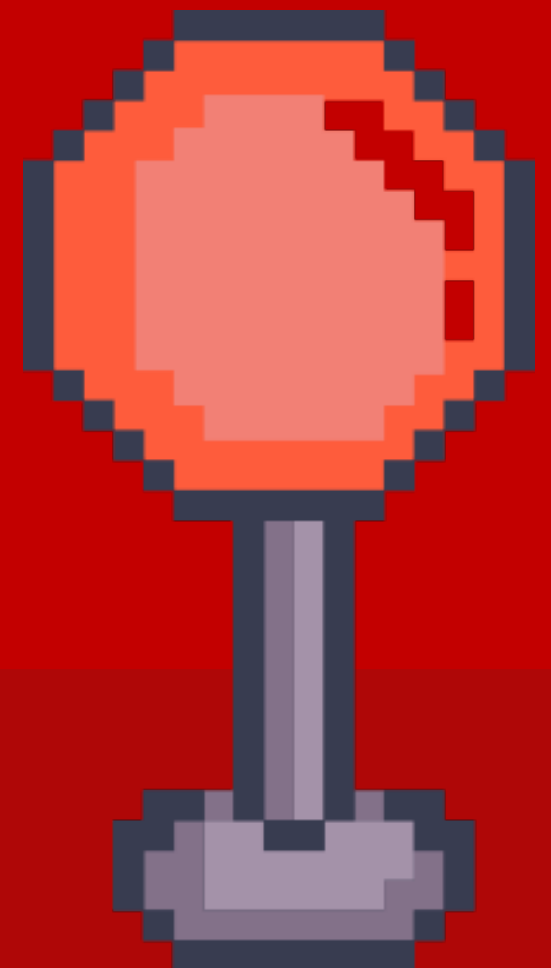
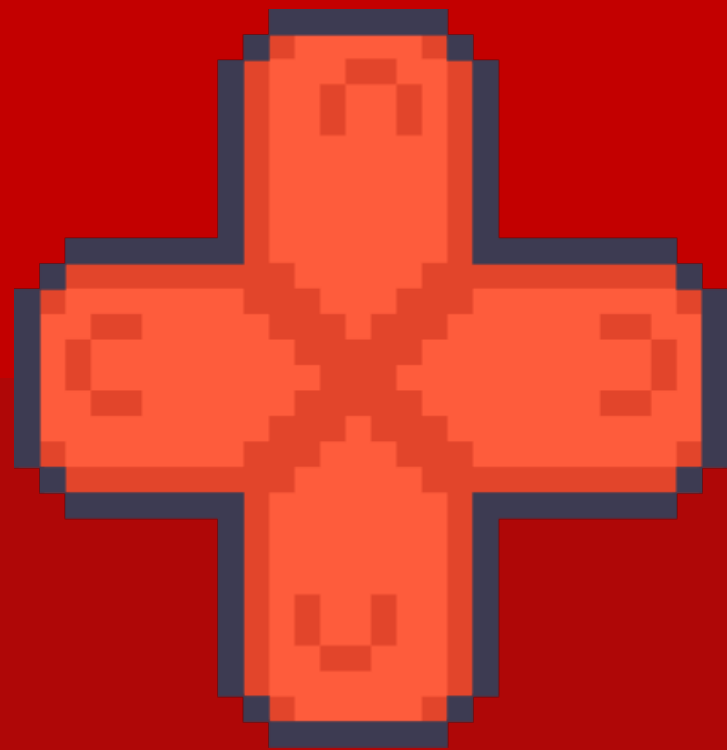
Table 3

Final Results on EXIST 2025 Test Set on Hard Label Evaluation on ICM-Hard

Subtask	English	Spanish	Overall	Overall Ranking
1.1 (Binary Sexism Detection)	0.6231	0.7124	0.6774	1
1.2 (Source Intention Detection)	0.3676	0.5937	0.4991	1
1.3 (Sexism Categorization)	0.5085	0.7650	0.6519	1



ERROR ANALYSIS



COMMON ERROR

Matrilocal husband? First time hearing this 😂
Googled it means the groom will stay at the bride's
parents' house. Opposed the common patrilocal
marriage. Anyway 1st ep is already interesting 😊
#MyHeroicHusband



Prediction

Sexist → Reported → Objectification



Gold Label

Non-sexist

COMMON ERROR ...

@radicalanthro No one is denying the existence of matriliney. But patrilocal clans were the dominant mode of social organisation for most of the world for the past 5000 years. And those patrilocal clans - in Europe, South Asia, the Middle East, and East Asia - organised to protect their assets



Prediction

Non-sexist



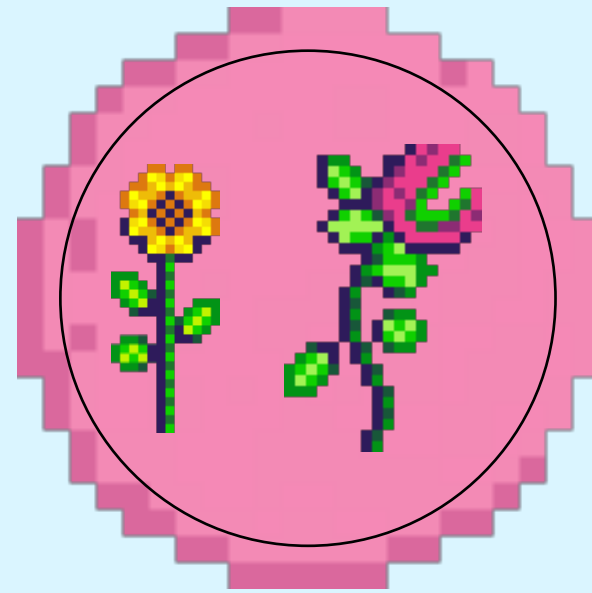
Gold Label

Sexist —> Reported --> Ideological-inequality

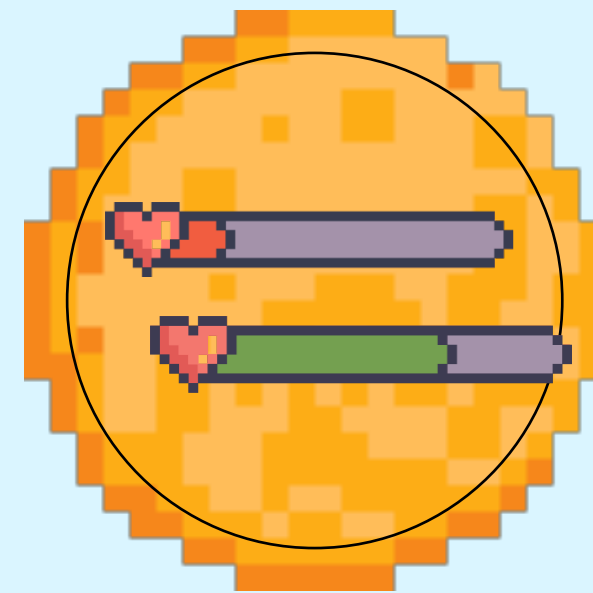
KEY TAKEAWAYS



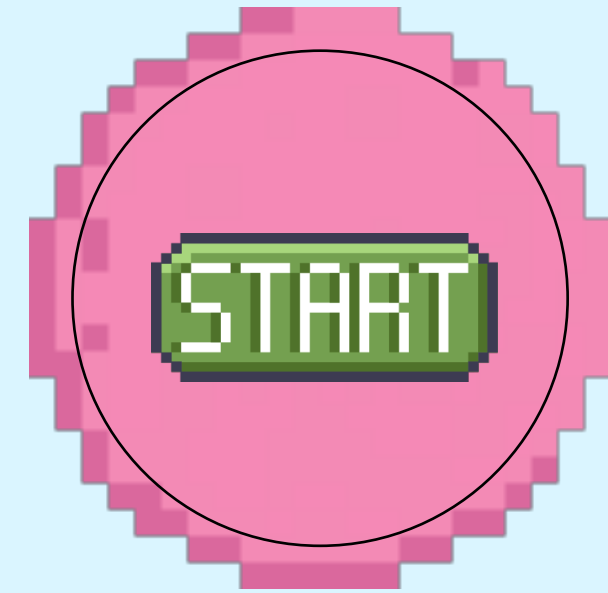
EASY APPROACH



MULTI-LINGUAL



EXPLICIT HIERARCHICAL
MODELING

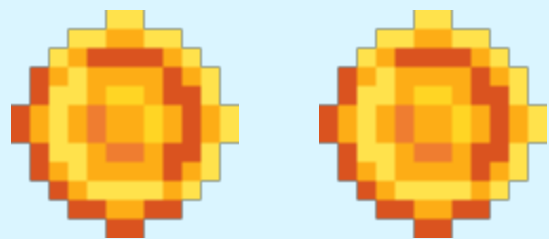
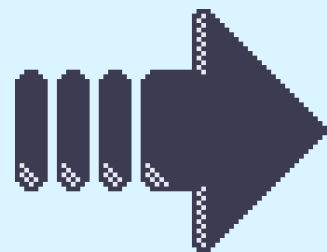


DELOYMENT





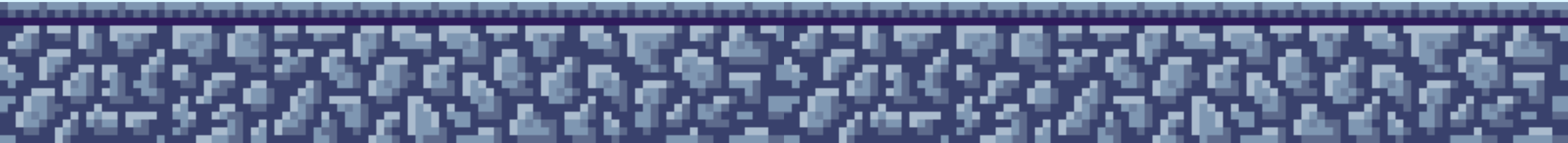
ADVANCE TO THE NEXT
LEVEL

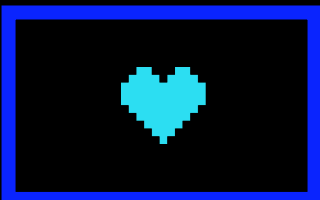
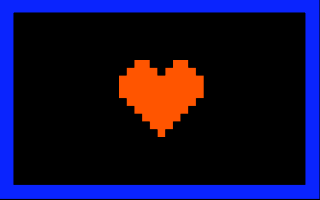


Persona based analysis
Demographic information



Multi-modal
Mario will try to tackle more tasks next time!





THANK
YOU

END

Table 1
 LoRA hyperparameters and training configuration

Configuration	Value/Setting
<i>LoRA Hyperparameters</i>	
Rank (r)	16
Alpha (α)	16
Target Modules	All linear layers (<code>all-linear</code>)
Dropout	0.1
Modules to Save	<code>lm_head, embed_tokens</code>
<i>Training Configuration</i>	
Quantization	4-bit QLoRA
Attention	Flash Attention v2
Learning Rate	2×10^{-4}
LR Schedule	Constant with 10% warmup
Batch Size	8 per device
Gradient Accumulation	2 steps
Gradient Checkpointing	Enabled